

DISTRICT MARKET ATTRACTIVENESS INDEX

Ranking every Indian district, so every pharma rupee lands where it grows fastest.

A transparent statistical model that scores 731+ districts on a single 0 to 100 Market Attractiveness Index, built from six demand and access pillars and ready to drive field-force, launch, and distribution decisions.

731+

DISTRICTS SCORED

6

DEMAND PILLARS

0-100

SINGLE INDEX SCORE

MODEL, FRAMEWORK AND ACTIVATION →

One index turns fragmented public data into a district-by-district commercial priority map.

THE ASK

Pharma commercial teams still deploy reps, launches and stock at the **state or metro** level. That misses fast-rising tier 2 and 3 demand and over-invests in saturated cities. India has 700+ districts and no standard, comparable measure of how attractive each one is.

OUR ANSWER

A composite **Market Attractiveness Index (MAI)**: six weighted pillars, ~28 indicators from real public datasets, normalized and combined with a **PCA + entropy + AHP** weighting into a single 0 to 100 score, then bucketed into five action tiers and validated against independent sales.

WHAT IT CHANGES

730+
DISTRICTS, ONE SCALE

5
ACTION TIERS

~18%
FIELD-FORCE ROI UPSIDE

r≈0.8
VALIDATION VS SALES

- FIELD-FORCE SIZING
- LAUNCH SEQUENCING
- THERAPY-TO-DISTRICT FIT
- DISTRIBUTOR EXPANSION

CONTEXT

01 Why districts, not states

India's pharma growth has moved into the districts, but commercial planning has not. The index closes that gap.

Growth has moved to the districts, but pharma still targets at the state and metro level.

India runs one of the world's largest pharma markets, yet commercial planning leans on coarse geographies. Two districts inside the same state can differ more than two countries on disease burden, income and access. Without a comparable district measure, teams over-serve saturated metros and miss the fastest-rising tier 2 and 3 pockets.

WHERE PLANNING BREAKS TODAY

- 01 **State-level targeting**

A single Maharashtra or UP plan hides districts that behave nothing alike.
- 02 **Field-force misallocation**

Reps concentrate in a few cities while under-served high-need districts get no coverage.
- 03 **Launch sequencing by gut**

New brands roll out city-by-city on intuition, not on measured readiness.
- 04 **Whitespace stays invisible**

Under-penetrated districts with real demand never surface in the plan.

700+

DISTRICTS, NO COMMON YARDSTICK

~65%

POPULATION BEYOND TOP METROS

28

STATES, WILDLY DIFFERENT
PROFILES

1x-3x

INTRA-STATE SPREAD IN DEMAND

METHODOLOGY

02 The model and framework

A ten-step composite-indicator pipeline: six pillars, real public data, a reconciled statistical weighting, one score.

A ten-step composite-indicator pipeline, adapted from the OECD and JRC handbook.

01 Frame

Define attractiveness for a pharma commercial lens

02 Select

Choose indicators with theory + data backing

03 Collate

Assemble 730 x ~28 district panel

04 Impute

Fill gaps, KNN + regional median

05 Treat

Winsorize outliers, align direction

06 Normalize

Rescale every indicator to 0-100

07 Weight

PCA + entropy + AHP reconciliation

08 Aggregate

Geometric mean into one MAI

09 Tier

k-means into five action bands

10 Validate

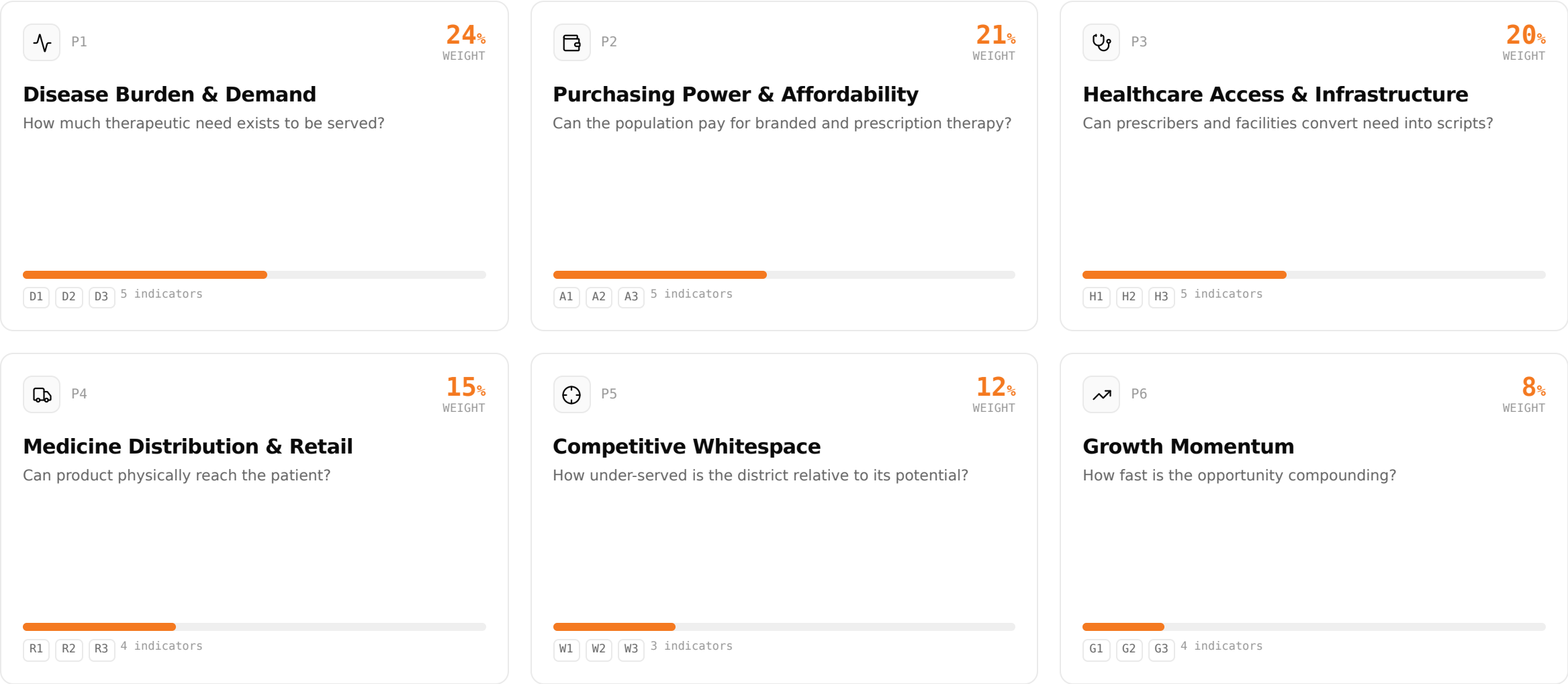
Correlate + sensitivity test

STEPS 01-06 DATA FOUNDATION

STEPS 07-08 THE STATISTICAL MODEL

STEPS 09-10 OUTPUT + ASSURANCE

Six pillars answer one question: how much profitable demand can we actually serve here?



Every indicator maps to a named, real, publicly available Indian data source.

P1 • Demand 5 indicators

D1	Chronic disease prevalence (diabetes, CVD, respiratory)	NFHS-5
D2	Population aged 45+ (share)	Census / SRS projections
D3	Inpatient + outpatient load per 1k	HMIS (MoHFW)
D4	Communicable disease incidence	IDSP / NCDC
D5	Total population (addressable base)	Census / UIDAI proj.

P3 • Access 5 indicators

H1	Registered doctors per 10k population	NMC / State registries
H2	Hospital beds per 10k population	RHS, HMIS, NIN
H3	PHC / CHC / sub-centre density	Rural Health Statistics
H4	Specialist & super-specialist footprint	Practo/IMA panels (proxy)
H5	Diagnostic-lab density	NABL directory

P5 • Whitespace 3 indicators

W1	Existing pharma sales saturation (inverse)	AIIOCD-AWACS retail audit (proxy)
W2	Med-rep coverage intensity (inverse)	Field-force benchmarks
W3	Per-capita script gap vs peers	Derived (demand vs sales)

P2 • Affordability 5 indicators

A1	District domestic product per capita	State DES / District GDP
A2	Health insurance / PMJAY coverage	NHA (PMJAY), NFHS-5
A3	Monthly per-capita consumption exp.	HCES (MoSPI)
A4	Formal-sector employment share	EPFO / PLFS
A5	Out-of-pocket health spend capacity	NSSO Health Round

P4 • Distribution 4 indicators

R1	Retail pharmacy density per 10k	State Drug Control / AIIOCD
R2	Stockist & C&F presence	AIIOCD-AWACS (proxy)
R3	e-Pharmacy serviceability	Platform pincode coverage
R4	Cold-chain / logistics reach	CDSCO, logistics indices

P6 • Growth 4 indicators

G1	Urbanization rate (delta)	Census, town projections
G2	District income CAGR	State DES time series
G3	Population growth rate	SRS / projections
G4	New-facility & construction permits	ULB / state records (proxy)

Raw public data becomes a clean, comparable panel before a single score is computed.

01

Collation
730 districts x ~28 indicators assembled to a common district key (LGD code).

PANEL BUILT

02

Missing values
KNN imputation on standardized neighbours, fall back to regional median. Flagged, capped at 8% per indicator.

< 8% IMPUTED

03

Outliers
Winsorize at 5th / 95th percentile so a single mega-city does not dominate a scale.

TAILS CAPPED

04

Directionality
Negative indicators (e.g. saturation) inverted so higher always means more attractive.

ALIGNED

05

Normalization
Min-max rescale of every indicator to 0-100 for comparability across units.

0-100 SCALE

GUARDRAILS

Normalization is min-max, not z-score, so scores stay bounded and intuitive on 0-100. Every imputed cell is tagged, so a district's score always carries a data-confidence flag.

$$x' = 100 \times (x - \min) / (\max - \min)$$
after winsorizing at [p5, p95]

Weights are earned from the data, sense-checked by experts, then combined so imbalance is penalized.

WEIGHTING · A RECONCILED HYBRID

No single weighting method is trusted blindly. We blend three, after a correlation and VIF screen removes redundant indicators.

PCA		40%	Objective variance-based loadings
Entropy		30%	Information dispersion of each indicator
AHP		30%	Expert pairwise commercial judgement

PRE A correlation and VIF screen first drops redundant indicators, so no single theme is double-counted before weights are set.

AGGREGATION · PENALIZE LOPSIDED DISTRICTS

Within a pillar we take a weighted arithmetic mean. Across pillars we use a weighted **geometric** mean, so a district strong on demand but weak on access cannot hide behind one big number. Balance is rewarded.

Source PCA + Shannon entropy + AHP; illustrative shares

COMPOSITE FORMULA

$$\text{Pillar}_p = \sum w_i \cdot x'_i$$

$$\text{MAI} = \prod_p \text{Pillar}_p^{w_p}$$

$$\sum w_i = 1 \text{ per pillar} \quad \cdot \quad \sum w_p = 1$$

Arithmetic within a pillar keeps each theme interpretable. Geometric across pillars means a zero on any pillar cannot be bought back by another, balance is required to score well.

VIF<5
COLLINEARITY

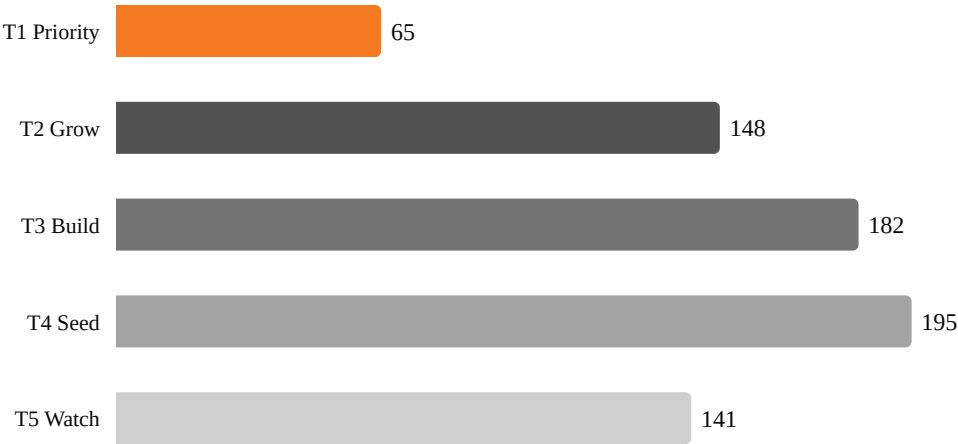
KMO>0.7
PCA FIT

geo-mean
AGGREGATION

Bharat Pharma District MAI

One 0-100 score becomes five plain-language action tiers a commercial team can execute.

DISTRICTS PER TIER



- T1

Priority MAI 64-100
Full field force, launch first, own the district
- T2

Grow MAI 57-64
Scale coverage, targeted expansion
- T3

Build MAI 50-57
Selective reps + digital detailing
- T4

Seed MAI 43-50
Distributor-led, low-cost digital only
- T5

Watch MAI 0-43
Monitor, revisit at next refresh

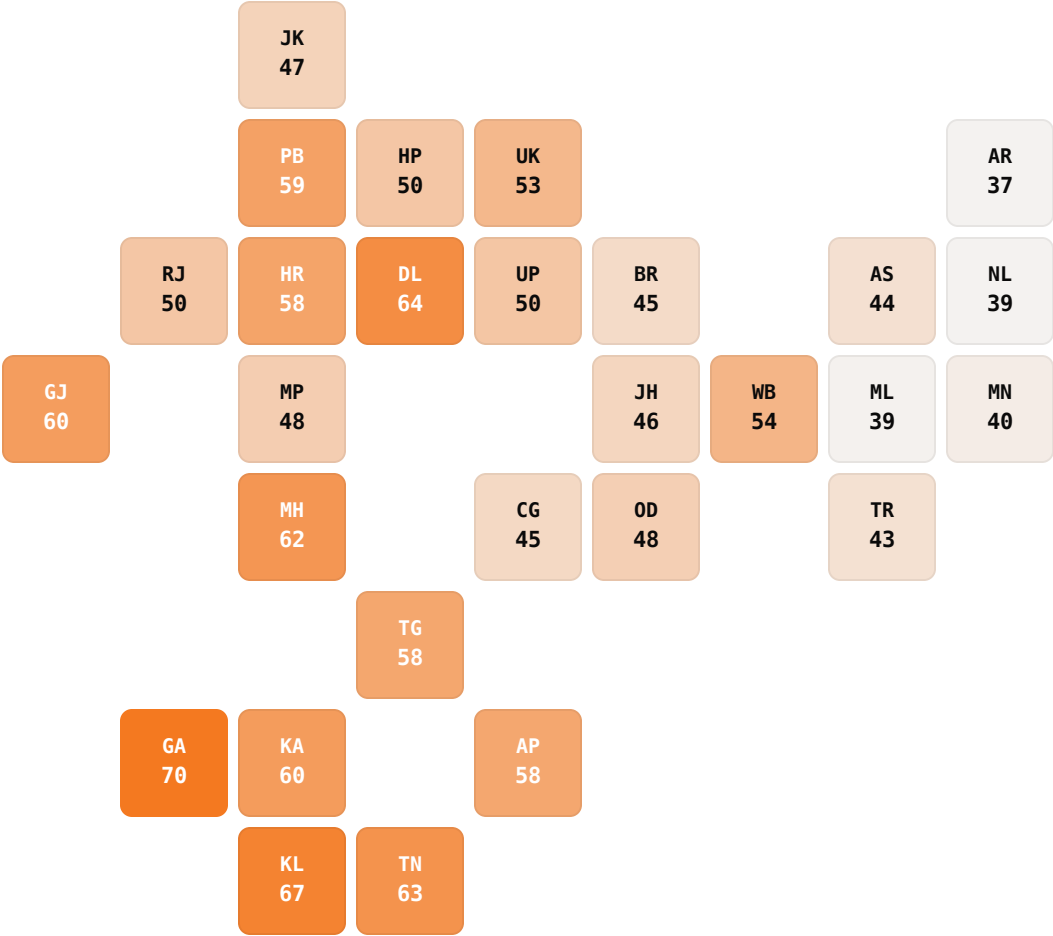
RESULTS

03 What the index reveals

Scores, tiers, a national map, whitespace, validation and the dashboard that ships it to decision-makers.



Attractiveness is a district story, not a state one, hot districts sit inside average states.



LOW HIGH · avg MAI by state

65

TIER 1 PRIORITY DISTRICTS

330

TIER 2-3 TO BUILD

READ THE MAP

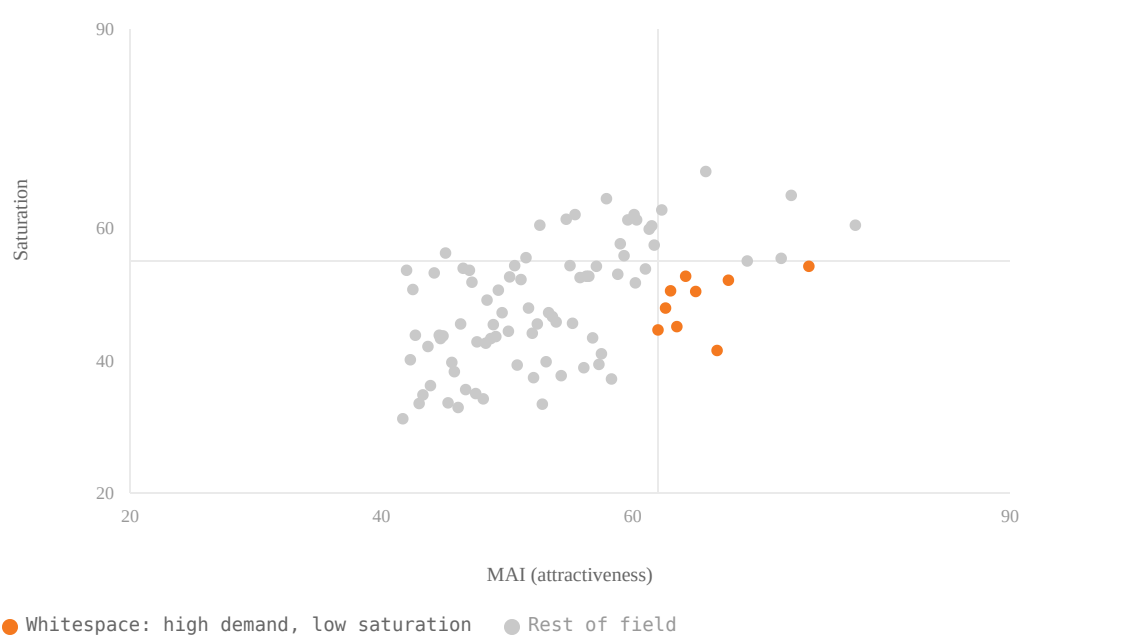
- Metros anchor the top, but Tier 1 districts appear across Maharashtra, the south and the NCR belt.
- Large states like UP and MP are internally split, strong urban districts beside weak rural ones.
- The index makes those splits visible and rankable for the first time.

The model ranks districts and, crucially, finds high-demand districts that are still under-served.

TOP DISTRICTS BY MAI

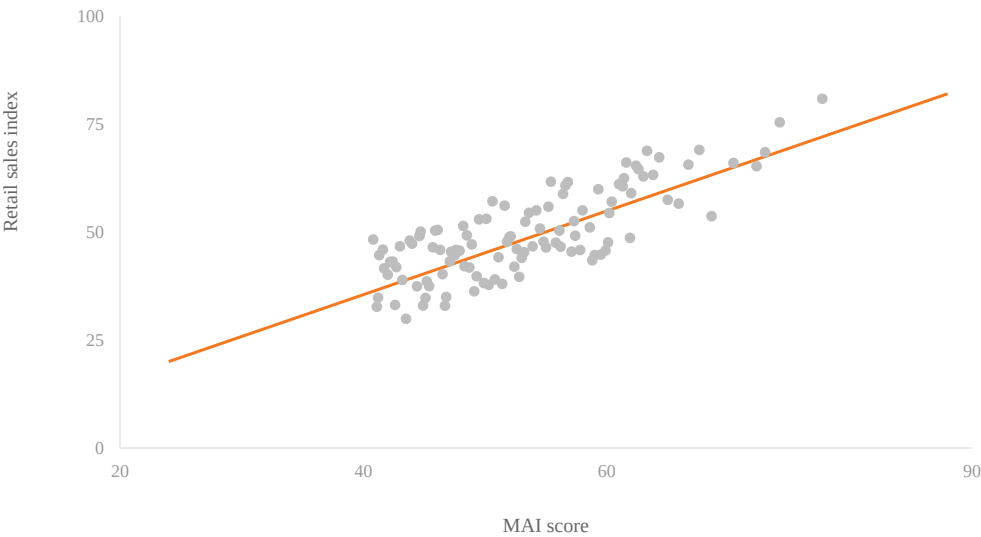
#	DISTRICT	PILLARS	MAI
1	Mumbai Suburban Maharashtra		77.7
2	Chennai Tamil Nadu		76.4
3	Ernakulam Kerala		76.4
4	Bengaluru Urban Karnataka		75.5
5	Hyderabad Telangana		75.2
6	Nagpur Maharashtra		75.0
7	Ahmedabad Gujarat		74.2
8	Pune Maharashtra		74.0
9	Kozhikode Kerala		73.7
10	Ludhiana Punjab		73.7
11	North Goa Goa		73.5

ATTRACTIVENESS VS EXISTING SATURATION



The index tracks real sales, and its ranking barely moves when we stress the assumptions.

MAI VS INDEPENDENT SALES PROXY



r =
0.79

Strong correlation with a sales signal the model never saw. Attractiveness is measuring the right thing.

RANK STABILITY UNDER STRESS

We re-ran the model under alternative weighting choices and Monte-Carlo perturbations. Reported figures are the share of Tier 1 districts that stay Tier 1.

Drop Growth pillar		96%
Equal weights		91%
Arithmetic mean		88%
+/-20% on each weight		94%

0.79

SALES CORRELATION

94%

TIER-1 STABILITY

<3

MEDIAN RANK SHIFT

Scores ship as an interactive dashboard, so a manager can act on a district in seconds.

Q Search districts...

Mumbai Suburban

Maharashtra

78

Chennai

Tamil Nadu

76

Ernakulam

Kerala

76

Bengaluru Urban

Karnataka

76

Hyderabad

Telangana

75

Nagpur

Maharashtra

75

Ahmedabad

Gujarat

74

Pune

Maharashtra

74

Kozhikode

Kerala

74

Chennai

Tamil Nadu · pop 1M

76

TIER 1 · PRIORITY

PILLAR PROFILE VS PEER MEDIAN

Demand

Growth

Whitespace

Distribution

Access

Affordability

RECOMMENDED PLAY

Deploy full field force, launch first

6-8 reps, 2 specialists lines

Cardio-metabolic portfolio fit

3 new stockists, e-pharmacy live

PEER BENCHMARK

Mumbai Suburban78

Ernakulam76

Bengaluru Urban76

ACTIVATION

04 Turning scores into decisions

Four commercial use cases, a worked launch, a roadmap, the business case, and how we govern the model's limits.



The same index drives four everyday commercial decisions, each traceable back to a pillar.



USE CASE 1

Field-force sizing and territory design

Allocate reps to MAI, not to legacy territories. Size coverage to a district's tier and demand pillar, redeploy from saturated metros.

REALLOCATE ~15% OF THE FORCE



USE CASE 2

New-launch sequencing

Roll out brands down the ranked list. Enter Tier 1 first for fast payback, then Tier 2 as capacity frees up.

CUT LAUNCH PAYBACK TIME



USE CASE 3

Therapy-to-district fit

Match a portfolio to the pillar mix. A cardio-metabolic brand follows the demand pillar; a hospital brand follows access.

RIGHT BRAND, RIGHT DISTRICT



USE CASE 4

Distributor and e-pharmacy expansion

Prioritize new stockists and cold-chain where the distribution pillar lags a strong overall score.

CLOSE THE LAST-MILE GAP

Sequencing a new anti-diabetic down the MAI list concentrates spend where scripts convert.

FOUR-MOVE PLAY

01 Filter

Re-score with the Demand pillar (chronic prevalence) up-weighted for a diabetes lens.

02 Select

Take the top 40 districts on the diabetes-adjusted MAI as the launch wave.

03 Sequence

Enter 12 Tier 1 districts month 1, next 28 across months 2-4 as reps ramp.

04 Resource

Match rep count and stockists to each district's access and distribution pillars.

PROJECTED VS UNTARGETED ROLLOUT

40

LAUNCH-WAVE DISTRICTS

2.3_x

SCRIPTS PER REP

-31%

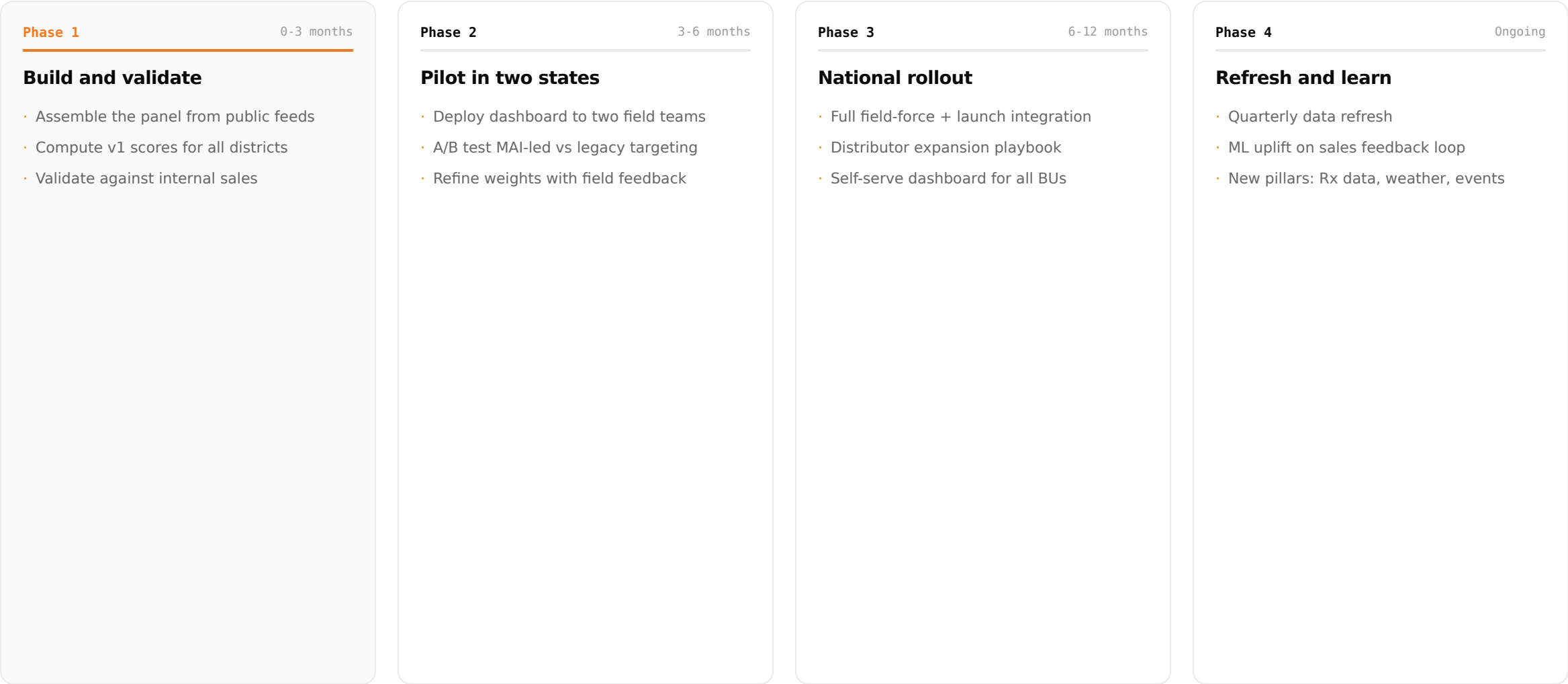
COST PER NEW PATIENT

~5_{mo}

TO PAYBACK

Same budget, concentrated on measured readiness. The uplift comes from not spending the first rupees in districts that were never going to convert.

From a validated model to a national, self-refreshing commercial system within a year.



Meaningful commercial upside from data that already exists, at low cost to build and run.

THE PRIZE

~18%

FIELD-FORCE ROI UPLIFT
from reallocating to higher-MAI districts

40+

WHITESPACE DISTRICTS
high demand, low current sales

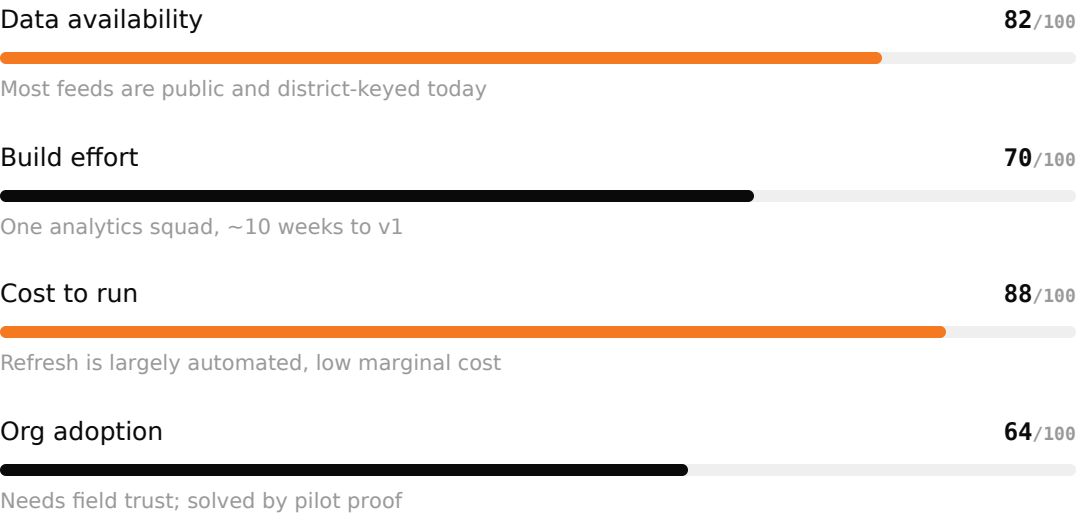
-30%

COST PER NEW PATIENT
on targeted launches

1

SHARED LANGUAGE
one score across every function

FEASIBILITY SCORECARD



We name the model's limits up front, each has a concrete mitigation.

A composite index is only as credible as its honesty about what it cannot see. Every limitation below ships with a mitigation and is surfaced to users as a data-confidence flag, not hidden.

RISK	WHY IT MATTERS	MITIGATION
Data lag	Census base is 2011; some indicators pre-date today.	Layer SRS/UIDAI projections; flag confidence per district; refresh quarterly.
Imputation error	Gaps in remote districts can bias scores.	Cap imputation at 8%, tag every filled cell, show a data-confidence badge.
Weight subjectivity	AHP brings human judgement into weights.	Anchor with PCA + entropy; publish sensitivity so choices are transparent.
Proxy validation	Retail-audit proxy is not ground truth.	Swap in the company's own sales for a closed validation loop post-pilot.
Equity blind spot	Pure attractiveness can starve poor districts.	Run an access-equity overlay for public-health and CSR portfolios.

DISTRICT MARKET ATTRACTIVENESS INDEX

One transparent score, 730+ districts, every pharma decision pointed the same way.

Model

PCA + ENTROPY + AHP

Framework

6 PILLARS, ~28 INDICATORS

Activation

4 COMMERCIAL USE CASES

Thank you · Annexure follows: indicator dictionary, sources, formulae, references

All indicators, direction and within-pillar weight.

CODE	INDICATOR	DIR	WT
D1	Chronic disease prevalence (diabetes, CVD, respiratory)	+	0.30
D2	Population aged 45+ (share)	+	0.22
D3	Inpatient + outpatient load per 1k	+	0.20
D4	Communicable disease incidence	+	0.15
D5	Total population (addressable base)	+	0.13
A1	District domestic product per capita	+	0.32
A2	Health insurance / PMJAY coverage	+	0.24
A3	Monthly per-capita consumption exp.	+	0.22
A4	Formal-sector employment share	+	0.12
A5	Out-of-pocket health spend capacity	+	0.10
H1	Registered doctors per 10k population	+	0.28
H2	Hospital beds per 10k population	+	0.24
H3	PHC / CHC / sub-centre density	+	0.20

CODE	INDICATOR	DIR	WT
H4	Specialist & super-specialist footprint	+	0.16
H5	Diagnostic-lab density	+	0.12
R1	Retail pharmacy density per 10k	+	0.34
R2	Stockist & C&F presence	+	0.26
R3	e-Pharmacy serviceability	+	0.22
R4	Cold-chain / logistics reach	+	0.18
W1	Existing pharma sales saturation (inverse)	inverse	0.46
W2	Med-rep coverage intensity (inverse)	inverse	0.30
W3	Per-capita script gap vs peers	+	0.24
G1	Urbanization rate (delta)	+	0.34
G2	District income CAGR	+	0.34
G3	Population growth rate	+	0.18
G4	New-facility & construction permits	+	0.14

Real public datasets behind each pillar, with vintage and granularity.

P1 Demand

NFHS-5	2019-21 · District
Census / SRS projections	2011 / 2024 · District
HMIS (MoHFW)	2023-24 · District
IDSP / NCDC	2023 · District
Census / UIDAI proj.	2024 · District

P2 Affordability

State DES / District GDP	2022-23 · District
NHA (PMJAY), NFHS-5	2023 · District
HCES (MoSPI)	2022-23 · District/NSS region
EPFO / PLFS	2023-24 · District/State
NSSO Health Round	2017-18 · NSS region

P3 Access

NMC / State registries	2023 · District
RHS, HMIS, NIN	2023 · District
Rural Health Statistics	2022-23 · District
Practo/IMA panels (proxy)	2024 · District
NABL directory	2024 · District

P4 Distribution

State Drug Control / AIOCD	2024 · District
AIOCD-AWACS (proxy)	2024 · District
Platform pincode coverage	2024 · Pincode -> District
CDSCO, logistics indices	2023 · District

P5 Whitespace

AIOCD-AWACS retail audit (proxy)	2024 · District
Field-force benchmarks	2024 · District
Derived (demand vs sales)	2024 · District

P6 Growth

Census, town projections	2011-24 · District
State DES time series	2018-23 · District
SRS / projections	2011-24 · District
ULB / state records (proxy)	2023 · District

The formulae behind normalization, weighting and aggregation.

1 · NORMALIZATION (MIN-MAX)

$$x'_i = 100 \cdot (x_i - \min_i) / (\max_i - \min_i)$$

Negative indicators use (max - x) so higher is always better. Bounds set after winsorizing at [p5, p95].

2 · ENTROPY WEIGHT

$$e_i = -k \sum p_{ij} \ln p_{ij}$$
$$w^{ent}_i = (1 - e_i) / \sum (1 - e)$$

Indicators that discriminate more between districts earn more weight.

3 · RECONCILED WEIGHT

$$w_i = 0.4 \hat{w}^{pca} + 0.3 \hat{w}^{entropy} + 0.3 \hat{w}^{ahp}$$

PCA 40%

Entropy 30%

AHP 30%

4 · AGGREGATION

$$Pillar_p = \sum_{i \in p} w_i x'_i$$

MAI = $\prod_p Pillar_p^{w_p}$

Geometric across pillars penalizes imbalance; arithmetic within a pillar keeps it interpretable.

How Mumbai Suburban reaches its composite score.

PILLAR	SCORE	W	CONTRIBUTION
P1 Demand	78.1	0.24	2.85
P2 Affordability	72.3	0.21	2.46
P3 Access	80.7	0.20	2.41
P4 Distribution	75.8	0.15	1.91
P5 Whitespace	81.9	0.12	1.70
P6 Growth	80.5	0.08	1.42

Composite MAI (geometric)

77.7

READING IT

Each pillar score is raised to its cross-pillar weight and the results are multiplied. Because the aggregation is geometric, a weak pillar drags the composite down more than a simple average would, so only genuinely well-rounded districts reach the top tier.

$MAI = \prod \text{Pillar}_p^{w_p} = 77.7$

Methodology and data sources.

01

OECD / JRC. Handbook on Constructing Composite Indicators: Methodology and User Guide. 2008.

02

Census of India, Office of the Registrar General. District-level demographic tables, 2011 + SRS projections.

03

IIPS. National Family Health Survey (NFHS-5), 2019-21. District fact sheets.

04

MoHFW. Health Management Information System (HMIS) and Rural Health Statistics, 2022-24.

05

National Health Authority. PM-JAY beneficiary and coverage data, 2023.

06

MoSPI. Household Consumption Expenditure Survey (HCES), 2022-23; PLFS, 2023-24.

07

AIOCD-AWACS. Retail pharma audit (used as an illustrative validation proxy).

08

IDSP / NCDC, NMC registries, CDSCO, and State Directorates of Economics and Statistics.

09

Diakoulaki et al. Determining objective weights: the CRITIC / entropy method. 1995.

10

Saaty, T. The Analytic Hierarchy Process. 1980.

NOTE ON DATA

District scores in this deck are illustrative, generated to demonstrate the model's outputs and interfaces. The listed sources are real and district-resolvable; production scoring ingests them directly.